

Binomischer Lehrsatz $(a+b)^n = \sum_{k=0}^n \binom{n}{k} \cdot a^{n-k} \cdot b^k$

↓
Pascal'sche Dreieck

$\Rightarrow$

$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$

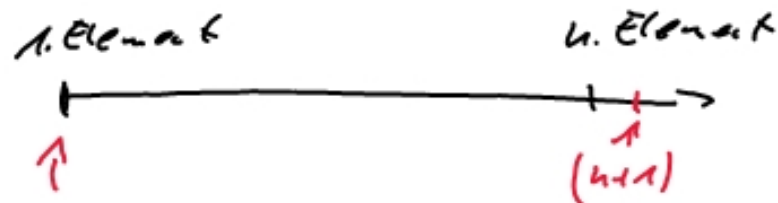
$\binom{4}{2} = \frac{4!}{2! \cdot (4-2)!} = \frac{4!}{2! \cdot 2!} = \frac{4 \cdot 3}{2}$

$(x+y)^4 = \binom{4}{0} x^4 y^0 + \binom{4}{1} x^3 y^1 + \binom{4}{2} x^2 y^2 + \binom{4}{3} x^1 y^3 + \binom{4}{4} x^0 y^4$

↓

$\frac{4!}{1! \cdot (4-1)!} = \frac{4!}{1! \cdot 3!} = 4$

Vollständig Induktion



$$\sum_{k=1}^n (2k-1) = 1 + 3 + 5 + 7 + \dots + (2n+1) = n^2$$

1. Anfang: $n=1$: $\sum_{k=1}^1 (2k-1) = 2 \cdot 1 - 1 = 1 = 1^2$ ✓

1. Schritt : Prämisse : Für alle $n \in \mathbb{N}^*$ gilt $\sum_{k=1}^n (2k-1) = n^2$

$n+1$ $\sum_{k=1}^{n+1} (2k-1) = (n+1)^2$ zu zeigen

$$\sum a_k = S_n$$

$$\sum_{k=1}^n (2k-1) + [2 \cdot (n+1) - 1]$$

$$n^2 + 2n + 1$$

$$(n+1)^2$$

$$3a) \sum_{k=1}^{n+1} \frac{1}{k \cdot (k+1)} = 1 - \frac{1}{(n+1)+1}$$

$$\underbrace{\hspace{1cm}}_{1 - \frac{1}{n+2}}$$

zu zeigen

$$\dots$$

$$1 - \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} = 1 + \frac{-1}{(n+1)} + \frac{1}{(n+1)(n+2)}$$

$$= 1 + \frac{-(n+2) + 1}{(n+1)(n+2)}$$

$$= 1 + \frac{-n-2+1}{(n+2)(n+1)}$$

$$= 1 + \frac{-n-1}{(n+2)(n+1)}$$

$$= 1 + \frac{-\overbrace{(n+1)}}{(n+2)\overbrace{(n+1)}} = 1 + \frac{-1}{n+2}$$